Farjan Rondón

Electrical Engineer | Power Electronics
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SUMMARY

Electrical Engineer with a broad base in power electronics, electrical machines, protection and power systems, built through coursework and hands-on laboratory work with transformers, induction motors, synchronous machines and inverters. Experienced in technical drawing with AutoCAD, motor protection studies in ETAP and circuit simulation, and in writing clear technical documentation in English. Motivated to apply this foundation to the electrical design of industrial equipment and prototypes.

EXPERIENCE

Jan 2025 – Feb 2026 — CARACAS, VENEZUELA

STUDY OF REAL-TIME JUNCTION TEMPERATURE ESTIMATION METHODS IN IGBTs AND POWER DIODES.
Undergraduate Researcher (Capstone Project).

- Studied the theory of junction temperature estimation from temperature-sensitive electrical parameters (TSEPs) and applied three methods experimentally: on-state voltage drop, di/dt during turn-off and internal gate resistance during turn-on.
- Designed and assembled the laboratory test setups around an inverter with six discrete IGBTs, and operated the measurement equipment.
- Calibrated each method by measuring its parameter at known temperatures under controlled operating conditions. This long measurement campaign produced the calibration curves used for estimation.
- Wrote MATLAB scripts to organize, process and plot all the calibration and test data, and to validate the results against the theoretical models.
- Designed an on-state voltage probe that measures only while the IGBT conducts: simulated it in LTspice, designed the PCB in KiCad, then built it and used it in the test setup.

Mar 2021 – May 2021 — REMOTE, MEXICO

UNIVERSIDAD ISEP
Web Developer

- Developed custom Odoo modules in Python to automate how lead data was captured, organized and stored.
- Fixed bugs, updated legacy code and wrote the technical documentation for each integration.

EDUCATION

2018 – 2026 — UNIVERSIDAD SIMÓN BOLÍVAR (USB) — CARACAS, VENEZUELA
Bachelor's degree in Electrical Engineering

- Academic program focused on power electronics, power systems, encompassing transmission line design, high-voltage engineering, electrical protection systems and energy conversion.
- Three-phase laboratory tests on transformers, induction motors and synchronous machines. High-Voltage laboratory tests and measurements with alternating, direct and impulse voltages, partial discharge and dielectric properties measurements, sparkover and potential distribution in insulator strings.

COURSES & CERTIFICATIONS

- Introduction to Power Electronic Converter Design – USB (March 2025).
- High-Voltage Installations and Insulation Studies – USB (March 2025).
- Motor and Critical Load Protection in Industry and Mining using ETAP – Greener Engineering School (March 2025).
- Earthing Systems: Measurement Methods – USB (March 2026).
- Data Analysis with Python – IBM (January 2021).
- Python Basics for Data Science – IBM (January 2021).

SKILLS

- **Simulation and analysis:** MATLAB (advanced), LTspice (advanced), PLECS (advanced), QuickField FEM (intermediate).
- **Hardware and CAD:** Schematic capture and PCB layout in KiCad (intermediate), AutoCAD (intermediate).
- **Laboratory:** Power electronics test setups, oscilloscope measurements, calibration, three-phase test on transformers and electrical machines.
- **Programming and data:** MATLAB scripting, Python, Excel.
- **Power systems:** ETAP (intermediate), DIgSILENT PowerFactory (basic).
- **Languages:** Spanish (native), English (fluent).